Nosh CV Intern Task — Report

Repo: github.com/maulik0jain/NoshInternTask · last updated 30 Oct 2026

Part	Status
Task 1 — ground-truth masks with a pretrained model	done
Task 2a — dev / eval split (4:1, leakage-aware)	done, frozen
Task 2b — classical image-processing segmentation + experiment log	done — 16 trials, 9 sweeps (§3)
Task 3 — evaluation & analysis	done — eval IoU 0.618 (§4)
Task 4 — video tracking (Drive link goes here)	next

Summary

- Ground truth: Grounding DINO (text "egg.") → boxes → SAM 2.1 masks, every overlay checked by eye
- Split: whole scenes go to one side (85 dev / 25 eval), checked for near-duplicates across the split
- Classical method: colour gate → **multi-threshold search for egg-shaped blobs** → GrabCut edge refinement → watershed split. No learning, all parameters picked on dev
- Eval (run once): IoU **0.618** (median 0.817), Dice 0.683, boundary F1 0.489; dev IoU 0.589
- Main gap: eggs on white / pale backgrounds and partly hidden eggs — the method relies on "the egg is the brightest egg-shaped thing"

0. Dataset — quick facts

- 110 images, all 640×640 (Roboflow resized with "stretch")
- Roboflow **folder** export → class labels only. **No boxes, no masks.**
- README says no augmentation → checked: 110 unique source ids, 0 copies
- 1–3 eggs per image, almost all white eggs
- Hard backgrounds: white bag, white egg trays, water in a steel pot, leaves, ash, hands, glossy lids

1. Task 1 — ground truth with pretrained models

Model choice

- SAM 2.1 needs a prompt (point / box) — it doesn't find eggs on its own
- Dataset has no boxes → **Grounding DINO** (open-vocabulary detector) turns the text `"egg."` into boxes
- Each box → **SAM 2.1 Hiera Base+** → mask
- Why not the lightest model: GT is made once, offline. Every GT mistake later counts as a mistake of my classical method → spend compute here
- Why not SAM "segment everything": it also segments hands, floor, shadows → messy filtering
- Also considered SAM 3 (text prompt built in) — heavier (~850M params) and gated weights; Grounding DINO + SAM 2.1 was enough

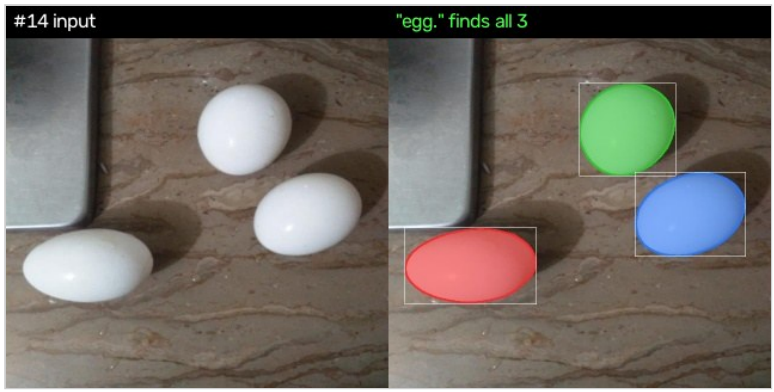
Pipeline

1. Grounding DINO base, prompt `"egg."`, box thr 0.30, text thr 0.25
2. Box cleanup: NMS (IoU 0.5), drop boxes >95% of the image, drop a "group" box that contains ≥2 other boxes
3. SAM 2.1, one mask per box (a box prompt isn't ambiguous → single-mask output)

4. Mask cleanup: clip to box (+6 px), fill holes (specular highlights), drop specks <15% of the biggest piece
5. Drop duplicate masks (>50% covered by a more confident egg)
6. Manual review overrides — 2 images, see below
7. Save binary mask, instance mask (ids left → right), overlay, `annotations.json`

Problems found while checking every overlay

Problem	Cause	Fix
Output files had wrong names	Filenames have extra dots (`x_jpg.rf.<hash>.jpg`), `with_suffix` cut off the hash	Build names from the stem
2 images missing one egg (#14, #35)	Prompt `"an egg."` gave those eggs 0.28–0.29 (threshold 0.30)	Prompt `"egg."`: real eggs 0.55–0.62, junk ≤ 0.10
Egg cut by a fridge-tray bar lost a piece (#24)	Cleanup kept only the largest piece	Keep all pieces ≥ 15% of the largest
Egg's reflection in a glossy lid detected as an egg (#61, #74)	Looks like a faint egg (det 0.36 / 0.47). Any rule for it would be fitted to 2 images	Manual override file `ground_truth_overrides.json`, with reasons
Not fixed: #17 — egg piece on the far side of a stem missing	SAM never segments that piece	Known GT error → shows up in Task 3



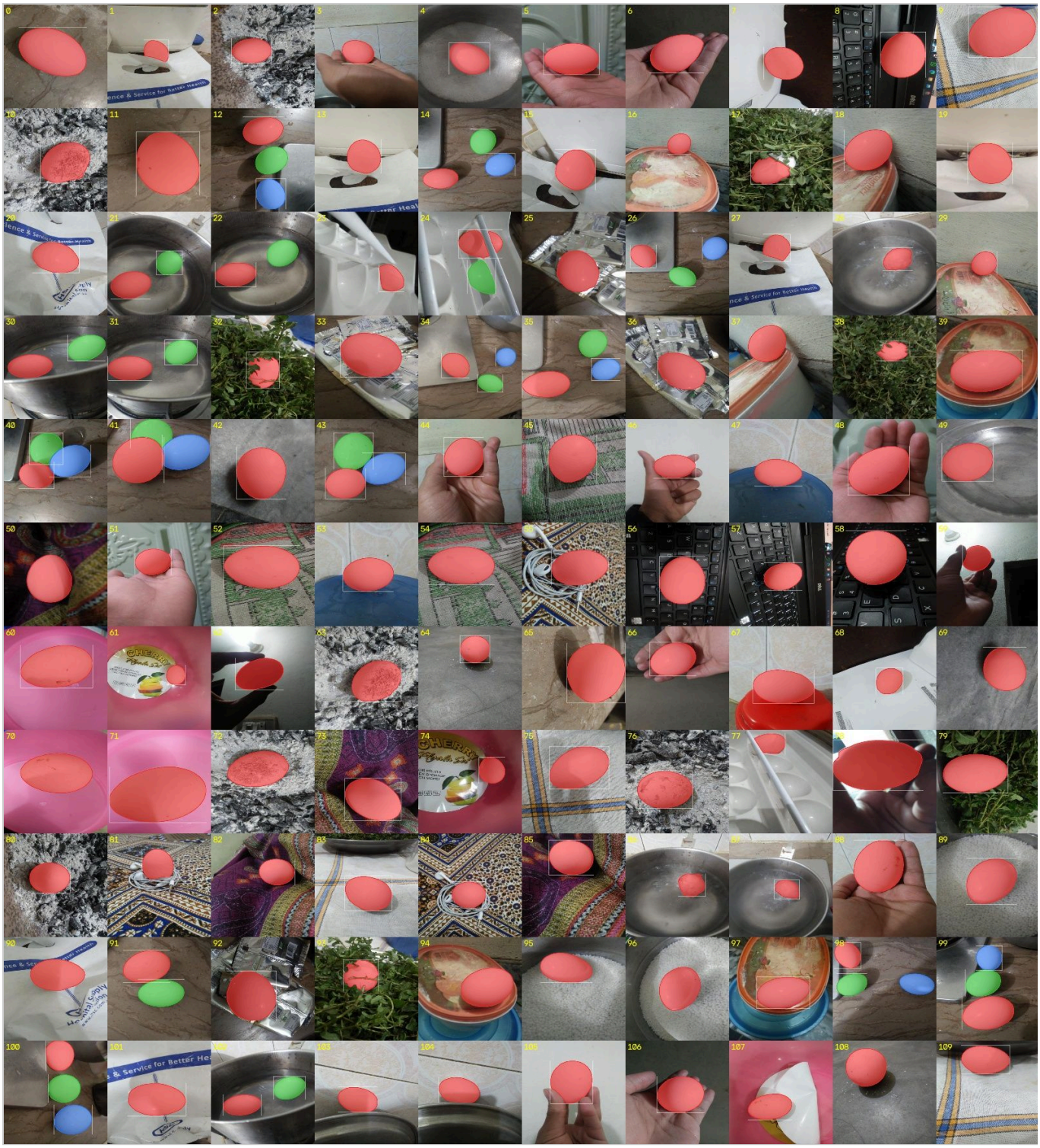
Prompt fix: all three eggs found with `"egg."`



Row 1: piece behind the bar kept. Row 2: reflection removed by override. Row 3: known GT limitation (not fixed).

Result

- **110 images, 139 eggs** — 92 images with 1 egg, 7 with 2, 11 with 3
- All 110 overlays checked by eye (full review sheet on the next page)
- GT = **visible** egg pixels. Leaves / fingers on top of an egg are not egg
- Known GT weak spots: eggs under leaves (#17), eggs partly hidden by tray bars



All 110 ground-truth overlays (colour = egg instance, white = detector box)

2. Task 2a — dev / eval split (4:1)

Why a random split is wrong here

- Photos were shot in sessions: same pot, same keyboard, same bag, same hand — different angles
- Random split → same scene in dev and eval → tune thresholds on a scene, then "test" on the same scene → eval looks better than it is
- Same idea as splitting by subject, not by frame

Steps

1. **Augmented copies**: strip `.rf.<hash>` → source id. 110 ids, 0 duplicates
2. **Near-identical shots**: pHash (64-bit DCT), Hamming ≤ 10 → 1 pair, same group
3. **Same scene**: DINOv2-small embeddings (CLS + mean patch), average-linkage clustering, cosine distance 0.30 → 35 groups
   - 0.30 picked by looking at the groups. 0.40 merged marble floor with grey floor (19 images) and still split the pot
4. **Manual merges (8)**: same physical setup the embedding kept apart — pot, keyboard, fridge tray, white bag + holder, pink bowl, painted lid, grey floor, towel + rug → `scene_merges.json` → **19 groups**
5. **Split whole groups**. Rules: eval = 17–23% of images; multi-egg share within ±5 points of overall; then as many scenes as possible. Seeded search (seed 42)
   - Try 1: `StratifiedGroupKFold` → valid, but eval had only 4 scenes
   - Try 2: soft penalty → 7 scenes, but only 1 multi-egg image in eval (too easy) → made the multi-egg share a hard rule

Result

	dev	eval
Images	85	25
Eggs	109	30
Scenes (groups)	14	5
Images with 2+ eggs	15%	20%

- Eval scenes: steel pot (12, incl. 5 multi-egg), painted container (7), dark-room hand (3), blue bowl (2), red bowl (1)
- 85 : 25 = 77 / 23, not exactly 80 / 20 — whole groups don't divide evenly

Leakage check

For each eval image: cosine similarity (DINOv2) to its most similar dev image.

Split	Mean of max similarity	Eval images with a near-twin in dev (> 0.9)
Random split, same size	0.80	16%
Group split (used)	**0.59**	**0%**

- pHash near-duplicate pairs across dev / eval: 0

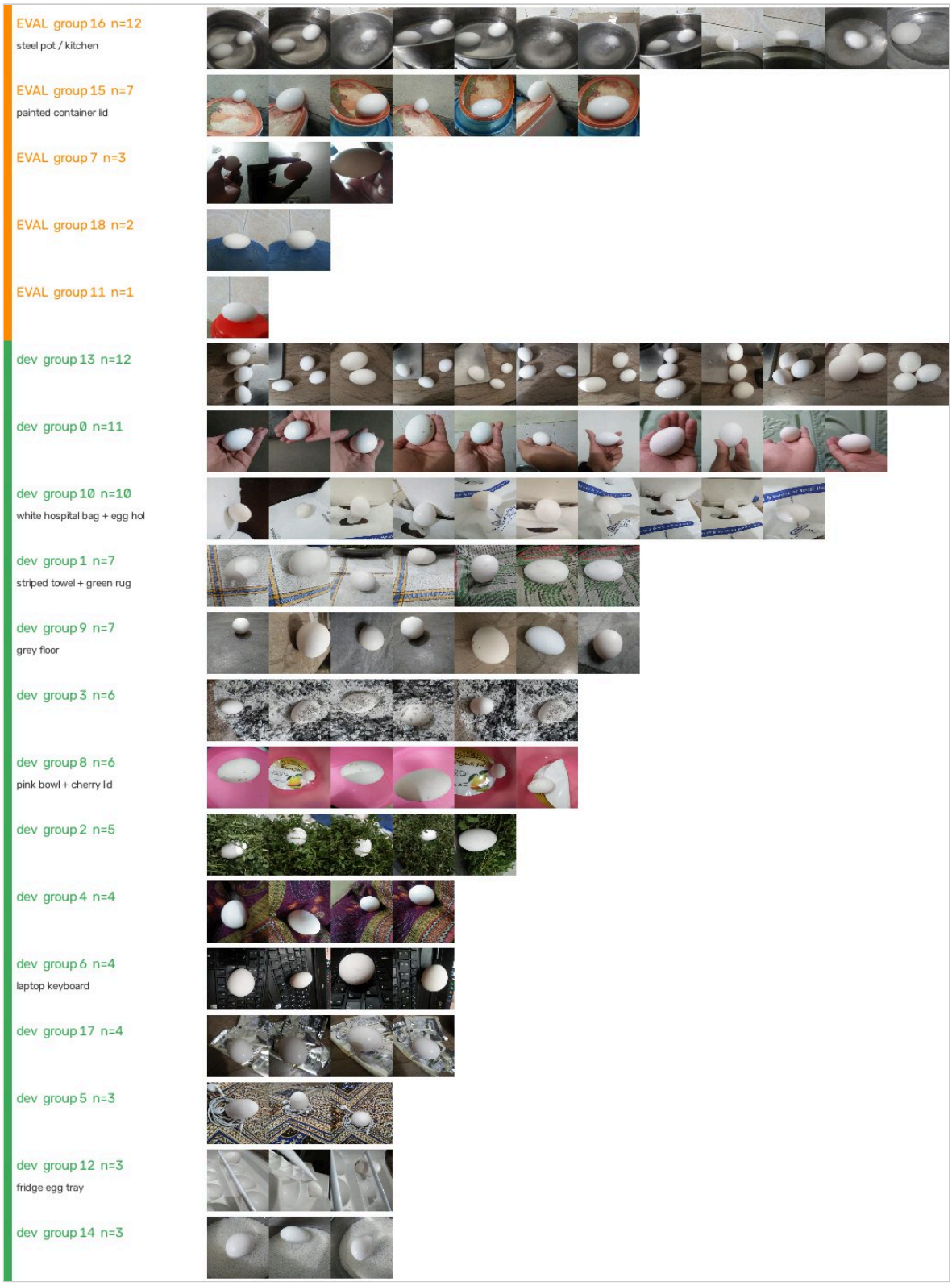
Rules from here on

- Split frozen in `data/splits/split.csv` (committed)

- All Task 2 tuning and experiments on **dev only** — `run_experiment.py` and `sweep.py` can't even load eval
- Eval touched **once**, by `final_eval.py`, which refuses to run a second time

Limitations

- Scene grouping is partly manual (my judgement) — every merge is listed in a file
- Some scene types are only in dev (leaves, ash, bright-room hands, marble-floor multi-egg) → eval numbers say less about those

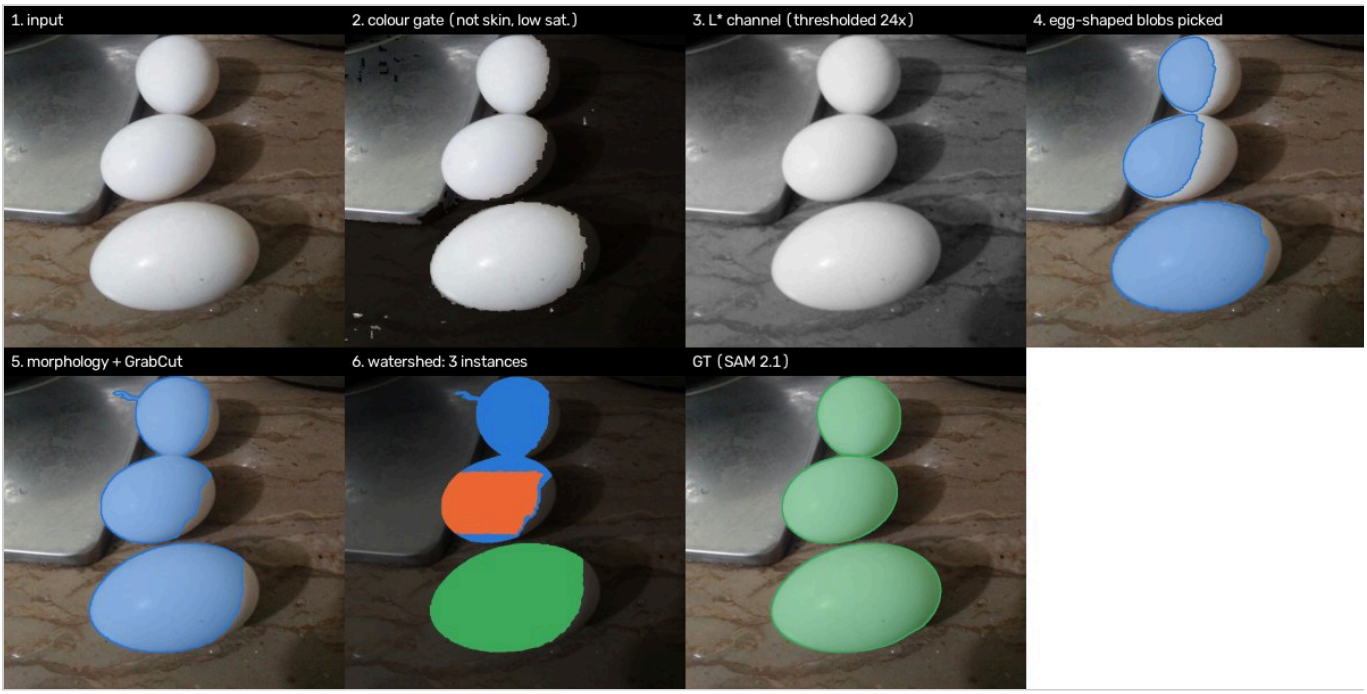


All 19 groups. Orange = eval, green = dev.

3. Task 2b — classical segmentation

Idea in one line

- An egg here = a **bright, almost colourless, egg-shaped** blob. No single threshold separates it from every background → try many thresholds, keep the blobs that look most like an ellipse, then fix their edges with GrabCut and split touching eggs with watershed



Final pipeline (T11) stage by stage on dev image #136 (three touching eggs). Step 4 keeps only the lit part of each egg; GrabCut (5) grows it to the shaded edge; watershed (6) separates the two eggs GrabCut merged.

Final algorithm (T11) — stages and parameters

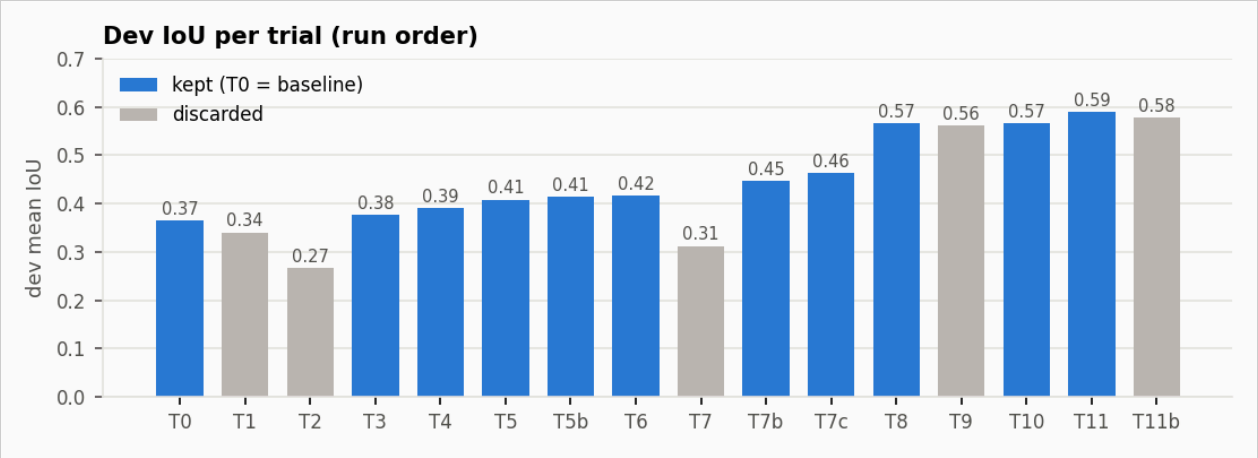
1. **Blur**: Gaussian 5×5 — shell texture and JPEG noise make ragged blob outlines
2. **Colour gate** — remove pixels that can't be egg before any thresholding
   - Skin: YCbCr box Cb 77–127, Cr 133–173 (Chai & Ngan 1999) → hands / fingers out. *Turned out to also remove warm-lit egg pixels — see the §4 ablation*
   - Saturation: HSV S ≤ 40 → coloured bowls, leaves, pink lids out (eggs are white / grey)
3. **Multi-threshold egg search** on the Lab L* channel (my own stage, similar idea to MSER)
   - 24 thresholds, evenly spaced between the 20th and 99.5th percentile of L* in that image
   - At each: `L* ≥ t` AND colour gate → opening 3 px (cuts thin bridges to background) → connected components → fill holes
   - Shape check per blob: area 0.003–0.6 of the image, solidity ≥ 0.85, axis ratio ≤ 2.2, ≤ 0.35 of the outline on the image border
   - **Egg score** = IoU(blob, its fitted ellipse) × solidity — an egg seen from above is ~1.0; background blobs are not. Keep score ≥ 0.8
   - Pick greedily, best score first. The same egg shows up at several thresholds (nested blobs) → take the **largest** one within 0.03 of the best score, so the egg isn't shrunk to its bright centre. Skip blobs that overlap a picked egg by > 0.2
   - Fallback: if nothing reaches 0.8, search once more with score ≥ 0.65 (only images that would otherwise be empty)
4. **Morphology**: opening 3 px, closing 7 px, fill holes (specular highlights inside eggs)

5. **GrabCut** (Rother 2004) on the original colours, seeded from the mask
   - Mask eroded by 10 px = sure egg · mask = probably egg · up to 70 px around it = probably background · further away = sure background
   - 5 iterations, fixed RNG seed; new blobs that don't touch the seed are dropped
   - Why: the search finds the lit part of the egg. The shaded side is often darker than parts of the background, so no threshold gets it — a colour model (GMM + graph cut) can
6. **Watershed** (Vincent & Soille 1991) on the distance transform of each blob: cores = pixels ≥ 0.6 × max distance (≥ 30 px) → one instance per core. Splits touching eggs (needed for Task 4)

- Output: binary mask + instance mask per image. 1.38 s / image on CPU (most of it GrabCut)
- Code: `src/eggseg/classical.py` (one small function per stage), parameters: `configs/default.yaml` + `experiments/T11/config.yaml`

How I ran experiments

- Each trial = a folder `experiments/Tn/` with `config.yaml`: which trial it builds on, what changed, why, decision + reason (written after the run)
- `run_experiment.py` runs it on the **85 dev images** → `results.json`, per-image CSV, worst-case panels. Small parameter sweeps with `sweep.py`, also dev only
- **Decision rule** (fixed before starting): keep a change if dev mean IoU goes up; if IoU ties, keep only if boundary F1 or instance F1 improves
- Metrics (same code as Task 3): IoU, Dice, pixel precision / recall, boundary F1 (2 px), instance F1 (match at IoU ≥ 0.5), number of empty predictions
- Table and per-trial notes below are generated from those files (`build_log.py` → `experiments/LOG.md`), not typed



Dev mean IoU of every trial in the order I ran them. Grey = discarded.

Experiment log — all trials (dev, 85 images)

Trial	Change	IoU mean	IoU med.	Prec.	Recall	Bnd F1	Inst F1	Empty	Decision
T0	Otsu threshold on grayscale (baseline)	0.366	0.273	0.407	0.879	0.158	0.008	0	baseline
T1	Otsu on CIE Lab L* instead of grayscale	0.340	0.256	0.373	0.896	0.143	0.007	0	discarded
T2	+ gray-world white balance and CLAHE on L*	0.266	0.206	0.282	0.875	0.077	0.005	0	discarded
T3	+ remove skin pixels (YCbCr box, Chai & Ngan 1999)	0.377	0.290	0.451	0.815	0.159	0.007	0	kept
T4	+ keep only low-saturation pixels (HSV S <= 60)	0.392	0.292	0.470	0.815	0.172	0.008	0	kept
T5	k-means (k=3) in Lab instead of Otsu; keep the brightest, least-coloured cluster	0.408	0.333	0.523	0.741	0.189	0.013	0	kept
T5b	k-means candidate + low-saturation filter (S <= 40, from the T4 sweep)	0.415	0.333	0.531	0.740	0.194	0.015	0	kept
T6	+ morphology (open r=3, close r=7, fill holes)	0.416	0.340	0.532	0.759	0.216	0.091	0	kept
T7	+ connected-component filters (area, solidity, axis ratio, border contact)	0.312	0.000	0.744	0.349	0.184	0.482	36	discarded
T7b	Multi-threshold egg search (24 thresholds, blobs scored by ellipse fit x solidity) replaces the single Otsu/k-means threshold	0.448	0.466	0.879	0.464	0.249	0.473	12	kept
T7c	Search on Lab L* instead of grayscale (from the T7b channel sweep)	0.464	0.483	0.902	0.483	0.257	0.493	11	kept
T8	+ GrabCut refinement seeded from the mask (eroded = sure egg, >70 px outside = sure background)	0.566	0.654	0.894	0.605	0.445	0.608	11	kept
T9	+ ellipse regularisation (blobs with ellipse IoU >= 0.9 are replaced by their fitted ellipse)	0.561	0.644	0.887	0.603	0.331	0.608	11	discarded
T10	+ distance-transform watershed to split touching eggs into separate instances	0.566	0.654	0.894	0.605	0.445	0.618	11	kept
T11	Fallback pass with a looser shape score (0.65) for images where the search found nothing	0.589	0.660	0.833	0.639	0.458	0.569	2	kept
T11b	Fallback pass returns at most 1 egg	0.579	0.660	0.821	0.618	0.456	0.603	2	discarded

How the experiments led to the algorithm

- **T0–T1: one global threshold.** Otsu on grey (0.366) finds eggs (recall 0.88) but also every bright background (precision 0.41). L* alone was worse
- **T2: global colour fixes hurt** (0.266) — gray-world and CLAHE brighten the background as much as the egg
- **T3–T5b: remove what can't be egg** — skin, then saturated colour, then k-means in Lab instead of Otsu. Small, steady gains (0.377 → 0.415), precision still poor
- **T6–T7: clean-up and shape filters.** Morphology was a tie. Shape filters on the single-threshold mask **broke** it (0.312, 36 empty masks): the egg is often glued to bright background, so filtering throws the egg away with it → the threshold itself is the problem
- **T7b–T7c: search over thresholds with an egg-shape score** — the key change. Precision jumps (0.53 → 0.88), IoU 0.464. Now the failure flips to under-segmentation (recall 0.48)
- **T8: GrabCut** fixes exactly that — biggest single gain (0.464 → 0.566, boundary F1 0.26 → 0.45)
- **T9: ellipse fitting** made edges worse (real eggs aren't perfect ellipses, partly hidden eggs even less) → discarded
- **T10: watershed** — no IoU change (as expected), instance F1 0.608 → 0.618

- **T11: fallback pass** for images with no egg found: empty masks 11 → 2, IoU 0.566 → 0.589 — but more false eggs (instance F1 0.569); T11b tried to limit that and lost IoU

Parameter sweeps (dev)

Sweep (dev)	On top of	Values → dev IoU mean	Picked
`watershed.peak_rel`	T10	0.4 → 0.566 · 0.5 → 0.566 · 0.6 → 0.566 · 0.7 → 0.566	0.6
`candidate.fallback_min_score`	T11	0.55 → 0.588 · 0.6 → 0.588 · 0.65 → 0.589 · 0.72 → 0.579	0.65
`saturation.s_max`	T4	25 → 0.395 · 40 → 0.395 · 60 → 0.392 · 90 → 0.383	25
`morphology.open`	T7	3 → 0.312 · 7 → 0.325 · 11 → 0.368 · 15 → 0.362	11
`candidate.channel`	T7b	"gray" → 0.448 · "L" → 0.464 · "whiteness" → 0.455	"L"
`candidate.min_score`	T7b	0.6 → 0.419 · 0.7 → 0.443 · 0.75 → 0.448 · 0.8 → 0.448	0.8
`saturation.s_max`	T7b	40 → 0.448 · 60 → 0.447 · 90 → 0.443 · 255 → 0.442	40
`grabcut.bg_dilate`	T8	15 → 0.515 · 25 → 0.536 · 45 → 0.559 · 70 → 0.566 · 100 → 0.563	70
`ellipse.min_fit_iou`	T9	0.9 → 0.561 · 0.95 → 0.565 · 0.97 → 0.565	0.95

Per-trial notes (what / why / params / result / decision)

T0 Otsu threshold on grayscale (baseline) `baseline`

- **Why:** Eggs are usually the brightest thing in the photo; Otsu picks the bright/dark split with no tuning.
- **Params:** on top of default → none (baseline)
- **Dev:** IoU 0.366 (median 0.273), boundary F1 0.158, instance F1 0.008, empty masks 0
- **Decision:** Reference point. High recall but low precision: Otsu takes every bright region, not just the egg.

T1 Otsu on CIE Lab L* instead of grayscale `discarded`

- **Why:** L* is perceptual lightness and ignores colour, so e.g. a saturated pink bowl is not "brighter" than it looks.
- **Params:** on top of T0 → `candidate: {"channel": "L"}`
- **Dev:** IoU 0.340 (median 0.256), boundary F1 0.143, instance F1 0.007, empty masks 0
- **Decision:** IoU fell (vs T0). With one global threshold, L* lets in even more bright background. Re-tested later inside the multi-threshold search (T7c), where it won.

T2 + gray-world white balance and CLAHE on L* `discarded`

- **Why:** Warm bulbs tint white eggs cream; CLAHE lifts eggs in dim/backlit photos.
- **Params:** on top of T1 → `preprocess: {"white_balance": true, "clahe": {"enabled": true, "clip": 2.0, "tile": 8}}`
- **Dev:** IoU 0.266 (median 0.206), boundary F1 0.077, instance F1 0.005, empty masks 0
- **Decision:** IoU fell sharply. Gray-world shifts colours in scenes dominated by one colour (pink bowl, green leaves), and CLAHE brightens the background as much as the egg, so Otsu took more background.

T3 + remove skin pixels (YCbCr box, Chai & Ngan 1999) `kept`

- **Why:** 14 images have a hand; a lit palm is as bright as the egg and gets swallowed by Otsu.
- **Params:** on top of T0 → `skin: {"enabled": true}`
- **Dev:** IoU 0.377 (median 0.290), boundary F1 0.159, instance F1 0.007, empty masks 0
- **Decision:** IoU and precision up vs T0; palms stop being swallowed with the egg.

T4 + keep only low-saturation pixels (HSV S <= 60) `kept`

- **Why:** Eggs are white/cream (near-colourless); many bright backgrounds are coloured (pink bowl, green leaves, wood, rugs).
- **Params:** on top of T3 → `saturation: {"enabled": true, "s_max": 60}`
- **Dev:** IoU 0.392 (median 0.292), boundary F1 0.172, instance F1 0.008, empty masks 0
- **Decision:** IoU up vs T3. Sweep T4_s_max: 25 and 40 tie, 40 used from T5b on because it is gentler on cream eggs.

T5 k-means (k=3) in Lab instead of Otsu; keep the brightest, least-coloured cluster `kept`

- **Why:** Otsu splits the histogram in two; with 3 clusters, egg / bright background / dark background can separate.
- **Params:** on top of T3 → `candidate: {"method": "kmeans", "k": 3, "chroma_weight": 1.0}`
- **Dev:** IoU 0.408 (median 0.333), boundary F1 0.189, instance F1 0.013, empty masks 0
- **Decision:** Better than Otsu (T4) as the candidate generator: IoU and precision up.

T5b k-means candidate + low-saturation filter (S <= 40, from the T4 sweep) `kept`

- **Why:** T4 and T5 helped separately; check whether they stack.
- **Params:** on top of T5 → `saturation: {"enabled": true, "s_max": 40}`
- **Dev:** IoU 0.415 (median 0.333), boundary F1 0.194, instance F1 0.015, empty masks 0
- **Decision:** T4 and T5 stack: IoU up again.

T6 + morphology (open r=3, close r=7, fill holes) kept

- **Why:** Raw masks are speckled (ash, rug texture) and have holes at specular highlights and dirt marks.
- **Params:** on top of T5b → `morphology: {"enabled": true, "open": 3, "close": 7, "fill_holes": true}`
- **Dev:** IoU 0.416 (median 0.340), boundary F1 0.216, instance F1 0.091, empty masks 0
- **Decision:** IoU about the same, boundary F1 and instance F1 clearly up (fewer specks, filled highlights).

T7 + connected-component filters (area, solidity, axis ratio, border contact) discarded

- **Why:** Precision is the problem (~0.5); most false positives are big, irregular or border-hugging bright background regions.
- **Params:** on top of T6 → `components: {"enabled": true}`
- **Dev:** IoU 0.312 (median 0.000), boundary F1 0.184, instance F1 0.482, empty masks 36
- **Decision:** IoU fell and 36 dev images got an empty mask: one global threshold glues the egg to bright background (ash, cables, white bag, towel), so the shape filter throws the egg away together with the background. Sweep T7_open: stronger opening helps but stays below T6.

T7b Multi-threshold egg search (24 thresholds, blobs scored by ellipse fit x solidity) replaces the single Otsu/k-means threshold kept

- **Why:** T7 failed because one global threshold glues the egg to bright background, so the shape filter throws both away (36 empty images). Trying many thresholds gives the egg a chance to separate at some level.
- **Params:** on top of T6 → `candidate: {"method": "search", "channel": "gray"}`, `components: {"enabled": false}`
- **Dev:** IoU 0.448 (median 0.466), boundary F1 0.249, instance F1 0.473, empty masks 12
- **Decision:** Best so far and precision up a lot; fixes the T7 failure mode. Sweeps: min_score 0.75 and 0.8 tie on mean IoU, 0.8 kept for its better median and instance F1; s_max 40 still best.

T7c Search on Lab L* instead of grayscale (from the T7b channel sweep) kept

- **Why:** T1 rejected L* with a single Otsu threshold; with shape-scored multi-threshold search the comparison can come out differently, so it was re-tested.
- **Params:** on top of T7b → `candidate: {"channel": "L"}`
- **Dev:** IoU 0.464 (median 0.483), boundary F1 0.257, instance F1 0.493, empty masks 11
- **Decision:** IoU up vs T7b: the channel sweep reverses the T1 result once candidates are chosen by shape.

T8 + GrabCut refinement seeded from the mask (eroded = sure egg, >70 px outside = sure background) kept

- **Why:** The search under-segments (precision ~0.88, recall ~0.48) - it often keeps only the bright cap of the egg. GrabCut's colour models can grow the mask out to the real egg edge.
- **Params:** on top of T7c → `grabcut: {"enabled": true, "iters": 5, "erode": 10, "bg_dilate": 70}`
- **Dev:** IoU 0.566 (median 0.654), boundary F1 0.445, instance F1 0.608, empty masks 11
- **Decision:** Largest single gain: GrabCut grows the egg cap out to its real edge (recall and boundary F1 up). bg_dilate 70 chosen by sweep T8_bg_dilate. Two code fixes found here: fixed RNG seed (runs were not repeatable) and drop GrabCut blobs that don't touch the seed (they were new background objects).

T9 + ellipse regularisation (blobs with ellipse IoU >= 0.9 are replaced by their fitted ellipse) discarded

- **Why:** Eggs are near-elliptical from most viewpoints; snapping clean blobs to the ellipse should fix dents from highlights/shadows on the outline. Irregular blobs (eggs behind fingers/leaves) are left alone because the GT only covers the visible part.
- **Params:** on top of T8 → `ellipse: {"enabled": true, "min_fit_iou": 0.9, "mode": "replace"}`
- **Dev:** IoU 0.561 (median 0.644), boundary F1 0.331, instance F1 0.608, empty masks 11
- **Decision:** IoU slightly down and boundary F1 down a lot: real egg outlines are egg-shaped (one blunt end), not ellipses, so snapping moves correct edges. Stricter fit (sweep T9_min_fit_iou) still loses.

T10 + distance-transform watershed to split touching eggs into separate instances kept

- **Why:** 5 images have touching eggs; a merged blob counts as zero correct instances. Binary IoU barely changes; instance F1 should.
- **Params:** on top of T8 → `watershed: {"enabled": true, "peak_rel": 0.6, "min_core_px": 30}`
- **Dev:** IoU 0.566 (median 0.654), boundary F1 0.445, instance F1 0.618, empty masks 11
- **Decision:** Binary IoU unchanged (as expected), instance F1 up: touching eggs now come out as separate objects. Needed for Task 4 tracking.

T11 Fallback pass with a looser shape score (0.65) for images where the search found nothing kept

- **Why:** 11 dev images still get an empty mask (IoU 0). The min_score sweep showed 0.6 leaves only 2 empty but hurts the other images, so the looser score is applied only where the strict pass found nothing.
- **Params:** on top of T10 → `candidate: {"fallback_min_score": 0.65}`
- **Dev:** IoU 0.589 (median 0.660), boundary F1 0.458, instance F1 0.569, empty masks 2
- **Decision:** IoU up and empty masks 11 -> 2 (the two left, #30 and #155, are white eggs in front of a white bag / white wall: at every threshold the egg merges with the background, so no egg-shaped blob exists). Cost: instance F1 down, because on hard images the looser pass also returns non-egg blobs. Sweep T11_fallback_min_score: 0.65 has the best IoU; 0.72 gives better instance F1 but loses IoU.

T11b Fallback pass returns at most 1 egg discarded

- **Why:** T11 removed 9 of 11 empty masks but its looser score also picked wrong blobs (precision and instance F1 down). An image that needs the fallback is a hard image; taking only its single best egg should keep the gain without the extra false eggs.
- **Params:** on top of T11 → `candidate: {"fallback_max_eggs": 1}`
- **Dev:** IoU 0.579 (median 0.660), boundary F1 0.456, instance F1 0.603, empty masks 2
- **Decision:** Instance F1 recovers most of what T11 lost, but IoU falls. Only 4 dev images change, all with 1 GT egg: on these hard images the best-scoring fallback blob is often not the egg (#118: the egg was one of T11's 3 blobs, the cap kept a background one). The score can't rank egg vs background here, so capping doesn't help. IoU first -> T11 stays final.

Things that didn't work (summary)

- **Global colour normalisation** (gray-world, CLAHE) — T2
- **Shape filtering after a single threshold** — T7: correct idea, wrong place. Moved into the threshold search (T7b)
- **Ellipse replacement** of the final mask — T9 (and the min-fit sweep: only harmless when it almost never fires)
- **Capping the fallback to 1 egg** — T11b: fewer false eggs, but on those hard images the best-scoring blob is often not the egg
- Bugs found on the way, each from looking at worst-case panels: GrabCut not repeatable (no RNG seed), GrabCut adding far-away background blobs, a degenerate `fitEllipse` result that tried to allocate 71 GB

4. Task 3 — evaluation & analysis

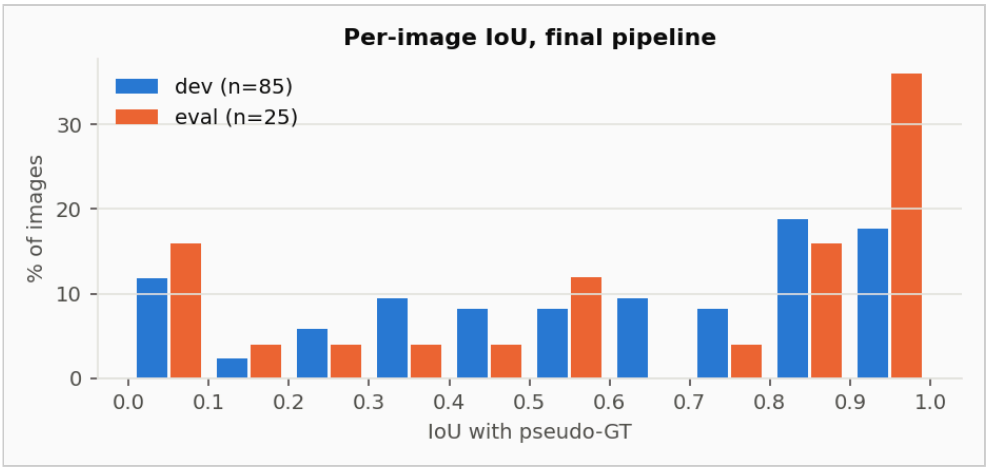
Setup

- GT = Task 1 SAM 2.1 masks, prediction = final pipeline **T11**, frozen before eval was opened
- `final_eval.py` ran dev + eval **once** (2026-10-30T15:17:18); `analyze.py` makes every number and plot below from its saved outputs
- Metrics and why:
  - IoU / Dice — overall overlap, the "match Task 1 quality" number. Per image, then averaged (an empty prediction counts as 0)
  - Pixel precision / recall — tells over- vs under-segmentation apart
  - Boundary F1 at 2 px — edge quality; IoU is forgiving on big eggs
  - Instance precision / recall / F1 (match at IoU ≥ 0.5) and exact egg count — what Task 4 tracking needs

Quantitative results

Metric (per-image mean unless noted)	dev (n=85)	eval (n=25)
IoU	0.589	0.618
IoU — median	0.660	0.817
Dice	0.680	0.683
Pixel precision	0.833	0.824
Pixel precision — median	0.990	0.994
Pixel recall	0.639	0.683
Boundary F1 (2 px)	0.458	0.489
Instance precision (IoU ≥ 0.5, pooled)	0.490	0.742
Instance recall (IoU ≥ 0.5, pooled)	0.679	0.767
Instance F1	0.569	0.754
Egg count exactly right (share of images)	0.671	0.800
Images with an empty prediction	2	2
Seconds per image (CPU)	1.52	1.38

- **Eval is a bit higher than dev** (IoU 0.618 vs 0.589). Not tuning luck — scene mix: the hardest backgrounds (white bag, ash, fridge tray, all IoU ≤ 0.34) are only in dev; eval's biggest scene is the steel pot (10 images, IoU 0.83)
- **All-or-nothing**: mean ≪ median. Eval: 13 / 25 images at IoU ≥ 0.8, 4 at exactly 0. Dev: 31 / 85 and 8. When the method finds the egg it's usually good; the mean is dragged down by misses
- **Under-segmentation, not false background**: pixel precision median 0.99 (eval), recall mean only 0.68 → what's predicted is almost always egg, but part of the egg is missing
- **Edges**: boundary F1 0.49 — even good masks are often more than 2 px off SAM's edge, mostly on the shaded rim
- **Instances**: eval instance F1 0.75, dev only 0.57 — dev has keyboards, ash and white bags where extra blobs pass as eggs (instance precision 0.49; e.g. dev #18: 8 predicted eggs for 1 real). Egg count exactly right: 80% of eval images



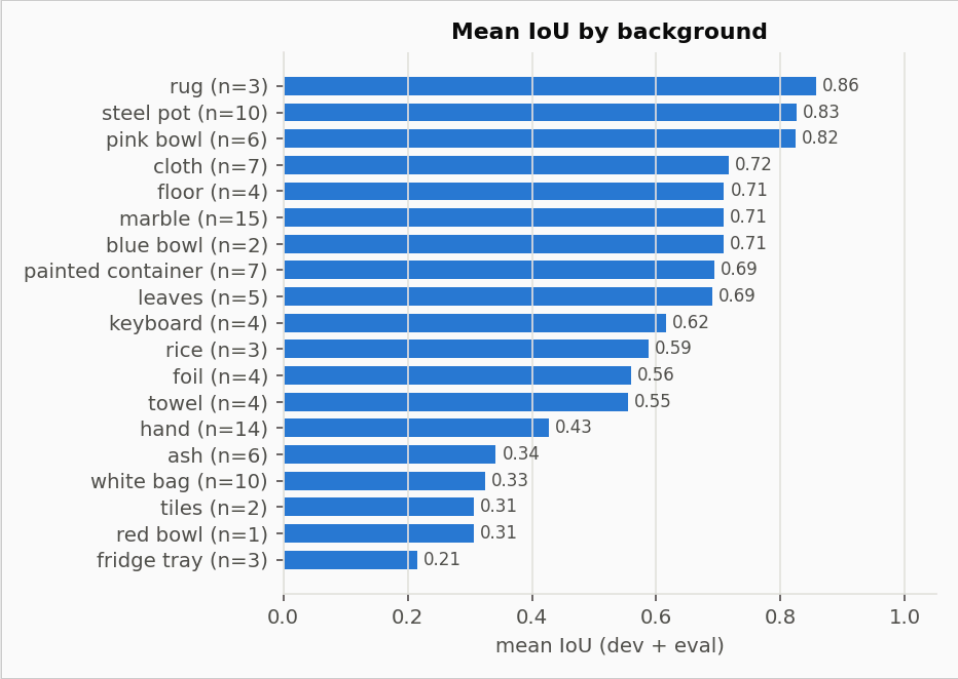
Per-image IoU, dev vs eval (share of images per 0.1 bin).

What the error depends on



Per-image IoU against number of eggs, image brightness and egg size. Spearman ρ (dev + eval): brightness -0.264, egg area 0.128.

- **Colour-gate loss is the strongest factor** (ρ = -0.336): in 12 images the skin / saturation gate throws away ≥ 40% of the egg before thresholding → mean IoU 0.29 vs 0.63 for the rest. Almost all of it is the **skin box**, not saturation (`diagnostics.csv`)
- **Hand in the image**: mean IoU 0.43 vs 0.62. On eval all 3 hand images score 0 — the dark-room / backlit shots
- **Brightness, egg size**: weak (ρ = -0.264 and 0.128). Bright images are often white backgrounds, hence the slight negative trend
- **Egg count**: multi-egg images do better (2 eggs 0.85, 3 eggs 0.71, 1 egg 0.56) — they were shot on high-contrast surfaces (steel pot, marble), so this is a scene effect, not a count effect



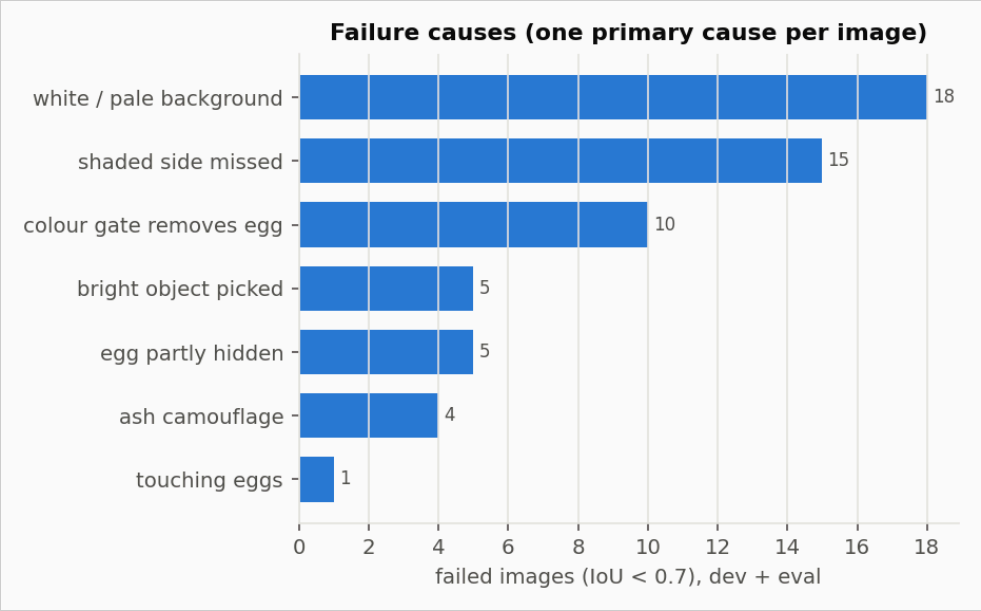
Mean IoU by background type (my manual labels in `data/processed/attributes_manual.csv`, dev + eval).

- Works: coloured or dark backgrounds — rug, steel pot, pink bowl (IoU ≥ 0.82)
- Fails: backgrounds as bright and colourless as the egg — fridge tray 0.21, white bag 0.33, pale tiles 0.31 — and ash 0.34 (egg dusted with the same grey). Red bowl is a single image
- That's the core assumption of the method breaking: "egg = brightest, least-coloured, egg-shaped thing"

Failure taxonomy

- Failure = IoU < 0.7 → 58 images (dev + eval)
- I looked at every one (sheets in `results/analysis/failures/`) and gave each **one primary cause** → `failure_labels.csv`

Primary cause	dev	eval	total
white / pale background	15	3	18
shaded side missed	12	3	15
colour gate removes egg	7	3	10
egg partly hidden	4	1	5
bright object picked	4	1	5
ash camouflage	4	0	4
touching eggs	1	0	1



Primary cause of each failed image.

- How labelled: (1) if most of the error is missed egg (FN > FP) and the colour gate removed ≥ 40% of the egg → "colour gate removes egg"; (2) otherwise by eye from the failure sheet. Notes per image in `failure_labels.csv`
- **White / pale background** (18) — no threshold separates egg from bag / tray / wall; the blob that comes out isn't egg-shaped → egg missed, or the bag picked instead
- **Shaded side missed** (15) — search keeps the lit half; the shaded half is darker than the surface next to it, and GrabCut only partly recovers it
- **Colour gate removes egg** (10) — the Chai & Ngan skin box also matches cream eggs, eggs under warm light, and backlit eggs. All 3 eval misses in the dark-room scene are this (84–98% of the egg removed)
- **Bright object picked** (5) — keyboard keys, foil, polished steel: bright, colourless and convex enough to pass the egg score
- **Egg partly hidden** (5) — leaves, tray bars, boiling water: the visible part isn't egg-shaped
- **Ash camouflage** (4) — egg dusted with ash, lying on ash: same grey, same texture; either nothing egg-shaped is found or the ash patch is taken with the egg
- **Touching eggs** (1) — the merged blob of 3 touching eggs fails the shape score before watershed ever sees it
- **Bad GT**: 0 — no failure was caused by a wrong SAM mask (the known GT error, #17, is not among the failures)

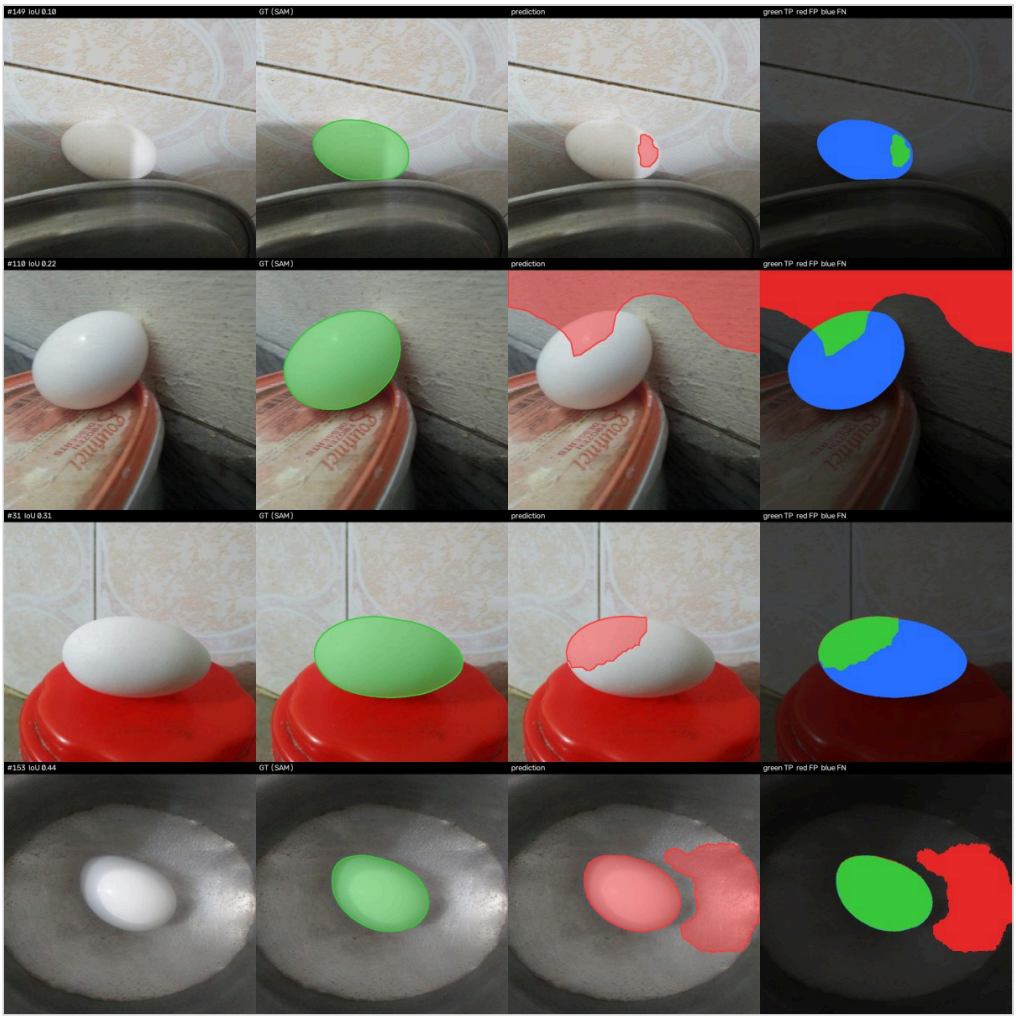
Worst eval images

- Columns: image · GT (SAM) · prediction · error map (**green** = correct egg, **red** = false egg, **blue** = missed egg)



Eval worst 1–4, rows top → bottom: #20, #24, #44, #139.

- **#20, #24, #44** — one scene: egg held up against a lamp / bright wall in a dark room. Backlit, the egg looks orange-brown → the skin box removes 84–98% of it → nothing egg-like is left (#44 keeps two finger blobs, #24 a wall switch, #20 nothing)
- **#139** — white egg against a pale wall: egg and wall merge at every threshold, so no blob is egg-shaped, even at the fallback score



Eval worst 5–8, rows top → bottom: #149, #110, #31, #153.

- **#149** — same as #139 with pale tiles: only a small highlight on the egg is kept
- **#110** — pale wall again; this time the wall itself is taken (big red area) together with a corner of the egg
- **#31** — lit top half found; the shaded lower half is not recovered by GrabCut
- **#153** — egg itself is right; a bright patch of steel / water next to it passes as a second egg

Ablation — what each stage still adds (dev, after the eval run)

- Final pipeline with one stage switched off at a time, **dev only** (`ablation.py`). Several stages were accepted early, on top of a much simpler method — do they still help?
- Run after the eval run, as part of this analysis → nothing here changes the reported pipeline or its eval numbers

Dev, one stage off	IoU mean	Δ IoU	Prec.	Recall	Bnd F1	Inst F1	Empty
final pipeline	0.589		0.833	0.639	0.458	0.569	2
no skin filter	0.609	+0.020	0.835	0.657	0.477	0.573	2
no saturation filter	0.587	-0.001	0.815	0.643	0.457	0.548	2
no morphology	0.580	-0.009	0.836	0.628	0.448	0.564	2
no GrabCut	0.483	-0.105	0.841	0.514	0.266	0.467	2
no fallback pass	0.566	-0.023	0.894	0.605	0.445	0.618	11
no watershed	0.589	+0.000	0.833	0.639	0.458	0.564	2

- **GrabCut** is the most important stage: without it IoU 0.483, boundary F1 0.266
- **Fallback pass**: removing it gives back exactly T10 (11 empty masks) → the IoU / instance-F1 trade-off from the log
- **Watershed**: same IoU by design, instance F1 0.564 → 0.569. Only 5 images have touching eggs
- **The skin filter now hurts**: without it dev IoU goes 0.589 → 0.609. It helped in T3, when the candidate was one Otsu threshold that grabbed whole hands. The shape search ignores hands anyway (not egg-shaped), so the skin box now mostly deletes warm-lit egg pixels — the same thing the colour-gate diagnostic and the failure taxonomy show. Lesson: re-check early decisions once the pipeline around them changes
- I did **not** switch it off in the final pipeline: that would be a change made after seeing eval results, and the eval run would no longer be a clean test. It's the first item below

Performance gaps → what I'd improve

1. **Colour gate** (10 failures + part of "shaded side"). Drop the skin box, or only remove skin regions that are large and touch the image border (hands come in from the edge). Dropping it alone: dev IoU 0.589 → 0.609 (ablation above)
2. **White / pale backgrounds** (18, the biggest group). Brightness and colour carry no signal there → use edges and shading: an egg is a smooth closed outline with a soft shading gradient and a cast shadow. E.g. ellipse detection on edges (Hough / RANSAC on Canny), or a gradient term in the egg score
3. **Shaded side missed** (15). The search finds the lit cap; give GrabCut the cap's fitted ellipse (slightly enlarged) as "probably egg". T9 forced the ellipse as the final mask, which was too strong; as a GrabCut seed the colour model still decides
4. **False eggs** on keys, foil, steel (5) and from the fallback pass. Add a check for straight edges / corners and high-frequency texture — eggs have neither
5. **Touching eggs**: split merged blobs with watershed *before* the shape score, so 2–3 touching eggs aren't rejected as one non-egg blob
6. **Evaluation itself**: GT is SAM pseudo-GT, so boundary F1 at 2 px also counts SAM's edge error; a few hand-drawn masks would show how much of the gap is the GT. Eval has only 5 scenes → per-scene numbers say more than the mean

- Bottom line: the method does well when the egg is the brightest, least-coloured, egg-shaped thing in view (IoU ≥ 0.8 on 13 / 25 eval images), and fails completely when that's not true. Closing the rest of the gap to SAM needs cues beyond brightness — edges, shading, context

5. Task 4 — video tracking

- Next. Google Drive link to the output video will go here

6. How to run

```
pip install -r requirements.txt
python scripts/download_assets.py          # SAM 2.1 checkpoint (+ dataset if missing)
python scripts/task1_sam2_groundtruth.py   # Task 1  -> ground_truth/
python scripts/make_split.py               # Task 2a -> data/splits/split.csv

# Task 2b: one trial / a sweep (dev only)
python scripts/run_experiment.py experiments/T11/config.yaml --worst 8
python scripts/sweep.py experiments/T8/config.yaml grabcut.bg_dilate "[25, 45, 70]" --name T8_bg_dilate
python scripts/build_log.py                # -> experiments/LOG.md, log.json, progress.png

# Task 3: the single eval run, then analysis + this report
python scripts/final_eval.py               # trial named in configs/final.yaml
python scripts/analyze.py                  # -> results/analysis/
python scripts/ablation.py                 # dev only -> results/ablation/
python scripts/build_report.py             # -> report/report.html + report.pdf
pytest tests/
```

- Tested on Windows 11, Python 3.13, PyTorch 2.6 + CUDA 12.4, RTX 4070 Laptop (8 GB). Task 2/3 need only OpenCV, NumPy, PyYAML, matplotlib (CPU)

7. AI tools

- Claude Code (AI coding assistant) used for: writing the scripts (GT pipeline, split, classical pipeline, experiment runner, evaluation, figures) and drafting this report
- Decisions (what to try next, keep / discard, failure labels) were made by looking at dev results and worst-case panels; every number in this report is filled in by `build_report.py` from saved result files

8. References

1. N. Otsu. A threshold selection method from gray-level histograms. IEEE Trans. SMC, 1979
2. D. Chai, K. N. Ngan. Face segmentation using skin-color map in videophone applications. IEEE Trans. CSVT, 1999
3. J. Matas et al. Robust wide-baseline stereo from maximally stable extremal regions (MSER). BMVC, 2002
4. A. Fitzgibbon, M. Pilu, R. Fisher. Direct least square fitting of ellipses. IEEE TPAMI, 1999
5. Y. Boykov, M.-P. Jolly. Interactive graph cuts for optimal boundary & region segmentation. ICCV, 2001
6. C. Rother, V. Kolmogorov, A. Blake. "GrabCut": interactive foreground extraction using iterated graph cuts. SIGGRAPH, 2004
7. L. Vincent, P. Soille. Watersheds in digital spaces: an efficient algorithm based on immersion simulations. IEEE TPAMI, 1991
8. S. Liu et al. Grounding DINO: marrying DINO with grounded pre-training for open-set object detection. 2023
9. N. Ravi et al. SAM 2: Segment anything in images and videos. 2024
10. M. Oquab et al. DINOv2: learning robust visual features without supervision. 2023